Huawei Access Node (BTM-ACC-34) Optical Transport Failure Troubleshooting and Resolution SOP

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8568

Description

The Huawei Access Node (BTM-ACC-34) is a critical component of the BTM Layout Aggregation site's optical transport network infrastructure. This SOP is designed to troubleshoot and resolve optical transport failures, specifically those related to SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, in the Access Node's role. The Huawei NE40E series of routers uses an Optical Transport Network (OTN) architecture, which is based on the ITU-T G.709 standard. The OTN architecture provides a robust and flexible framework for transporting client signals over optical fibers. However, it is not immune to failures caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. When these failures occur, they can cause critical alarms, such as ALARM_8568, which indicate a serious issue with the optical transport network. To troubleshoot and resolve these failures, this SOP will guide the network administrator through a series of diagnostic commands, resolution steps, and verification commands, ensuring that the network is restored to a stable and operational state.

Pre-requisite Checks

- check: Verify that the site is accessible for remote hands and that the necessary equipment and tools are available., condition: Ensure that the redundant link status is functioning correctly and that the network can be safely isolated for troubleshooting purposes.

- check: Check the site's environmental conditions, including temperature, humidity, and power supply., condition: Verify that the site's power supply is stable and that there are no issues with the power distribution units (PDUs).

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface brief, Expected Output: Interface Name Current State Current Status Last Error, Description: This command displays a brief summary of the optical transport interfaces, including their current state, status, and last error. This information is critical in identifying potential issues with the optical transport network.
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```
Command: display optical-transport sfp info, Expected Output: SFP Type SFP Serial Number SFP Status Current Temperature, Description: This command displays detailed information about the SFPs installed in the Access Node, including their type, serial number, status, and current temperature. This information is useful in identifying potential issues with the SFPs, such as degradation or overheating.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: replace optical-transport interface interface-name, Description: Replace the faulty SFP/XFP module with a new one. Ensure that the new module is properly installed and configured before proceeding.

Step 2: Step: Perform a fiber cut repair, Command: repair optical-transport fiber interface-name, Description: Perform a fiber cut repair by splicing or replacing the damaged fiber. Ensure that the repair is done correctly to avoid further issues with the optical transport network.

Post-Check Verification

```
Command: display optical-transport interface brief, Expected Output: Interface Name Current State Current Status Last Error, Description: Verify that the optical transport interfaces are functioning correctly and that there are no issues with the optical transport network.

Command: display optical-transport sfp info, Expected Output: SFP Type SFP Serial Number SFP Status Current Temperature, Description: Verify that the SFPs are functioning correctly and that there are no issues with the SFPs, such as degradation or overheating.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore optical-transport interface interface-name, Description: Restore the original SFP/XFP module to its original state. Ensure that the original module is properly installed and configured before proceeding.

- Step: Perform a fiber repair, Command: repair optical-transport fiber interface-name, Description: Perform a fiber repair by splicing or replacing the damaged fiber. Ensure that the repair is done correctly to avoid further issues with the optical transport network.